

Comparison between *E* and *M* (1)

	Electricity		Magnetism	
Physical Quantities	constant	$\frac{1}{4\pi\epsilon_0}$	constant	$\frac{\mu_0}{4\pi}$
	charge	q	current	i
	E-field	$\vec{E}$	B-field	$\vec{B}$
	E-potential	V	B-potential	To be continued
	capacity	C	inductance	L
	E-dipole	$\vec{p}$	B-dipole	$\vec{\mu}$

E means electric field; B means magnetic field

Comparison between E and M (2)

	Electricity		Magnetism	
Force	$\vec{F} = q\vec{E}$		$\vec{F} = q(\vec{v} \times \vec{B})$	
Field	Coulomb	$d\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \hat{r}$	Biot-Savart	$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{id\vec{s} \times \hat{r}}{r^2}$
	E-Gauss <i>Maxwell 1</i>	$\oiint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$	B-Gauss <i>Maxwell 2</i>	$\oiint \vec{B} \cdot d\vec{A} = 0$
	Faraday <i>Maxwell 3</i>	$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$	Ampere <i>Maxwell 4</i>	$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$
	E-Gauss <i>infinite rod</i>	$E = \frac{\lambda}{2\pi\epsilon_0 r}$	B-Gauss <i>infinite wire</i>	$B = \frac{\mu_0 i}{2\pi r}$
	E-Gauss <i>infinite plane</i>	$E = \frac{\sigma}{2\epsilon_0}$	B-Gauss <i>infinite sheet</i>	$B = \frac{\mu_0 J}{2}$

Comparison between E and M (3)

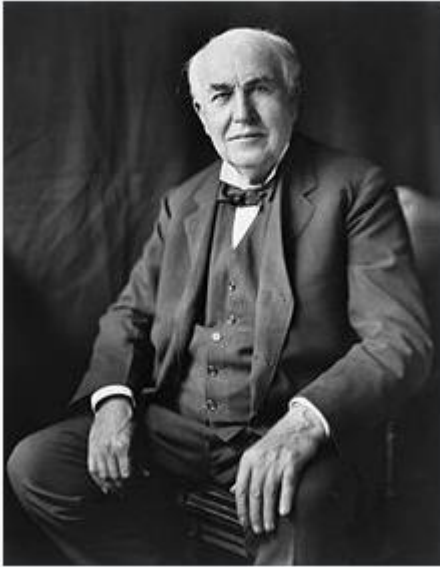
	Electricity		Magnetism	
Field and Potential	$\Delta V = - \int \vec{E} \cdot d\vec{s}$		To be continued	
Energy	capacitor	$C = \frac{q}{V}$	inductor	$L = \frac{N\Phi_B}{i}$
	capacitor <i>parallel plate</i>	$C = \frac{\epsilon_0 A}{d}$	Inductor <i>solenoid</i>	$L = \mu_0 n^2 l A$
	E-energy	$U_E = \frac{Q^2}{2C}$	B-energy	$U_B = \frac{Li^2}{2}$
	E-energy density	$\mu_E = \frac{1}{2} \epsilon_0 E^2$	B-energy density	$\mu_B = \frac{1}{2\mu_0} B^2$

Comparison between E and M (4)

	Electricity		Magnetism	
Dipole	E-dipole	$\vec{p} = q\vec{d}$	B-dipole	$\vec{\mu} = i\vec{A}$
	torque	$\vec{\tau} = \vec{p} \times \vec{E}$	torque	$\vec{\tau} = \vec{\mu} \times \vec{B}$
	potential energy	$U = -\vec{p} \cdot \vec{E}$	potential energy	$U = -\vec{\mu} \cdot \vec{B}$
	E-field <i>large z</i>	$\vec{E} = \frac{1}{2\pi\epsilon_0} \cdot \frac{\vec{p}}{z^3}$	B-field <i>large z</i>	$\vec{B} = \frac{\mu_0}{2\pi} \cdot \frac{\vec{\mu}}{z^3}$

Ch31 EM oscillation & AC current

Thomas Edison



Thomas Alva Edison (1847 – 1931) was an American inventor and businessman, who has been described as **America's greatest inventor.**

J. P. Morgan



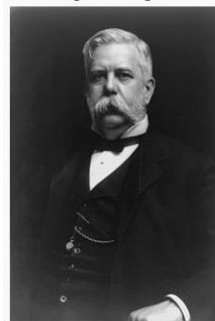
General Electric Company



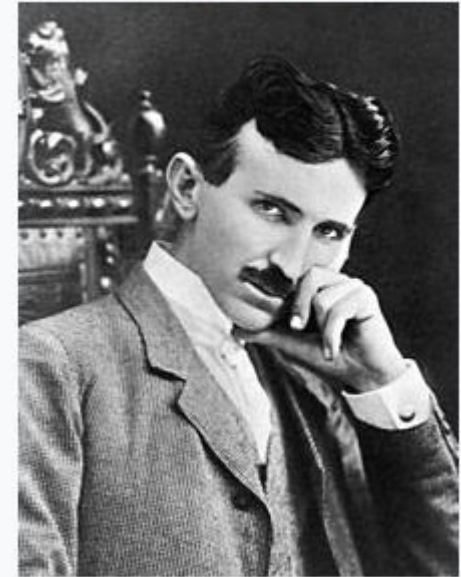
Westinghouse Electric Corporation



George Westinghouse



Nikola Tesla

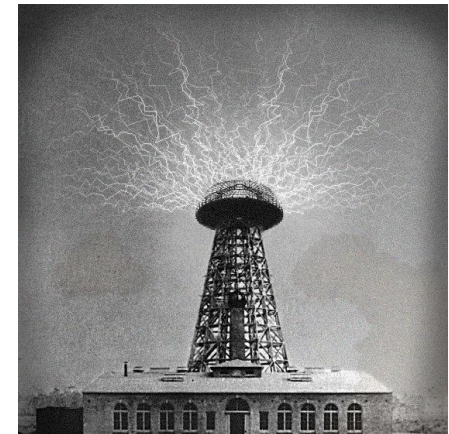
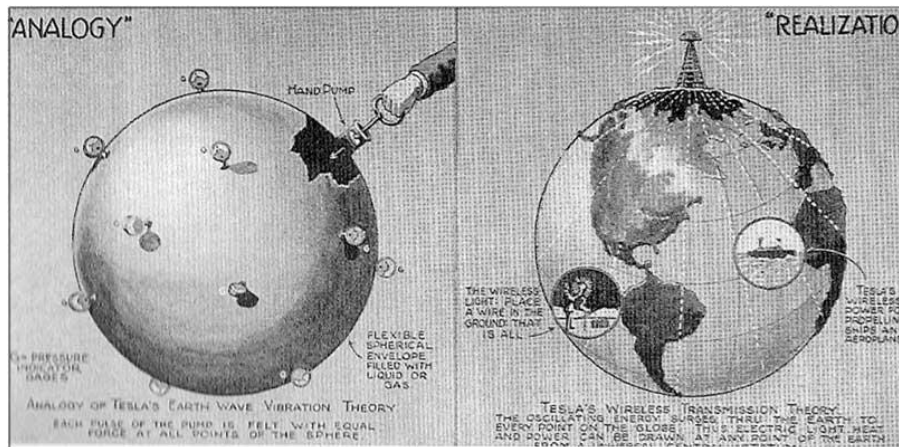


Nikola Tesla (1856 – 1943) was a Serbian-American inventor, electrical engineer, mechanical engineer, physicist, and futurist who is best known for his contributions to the design of the **modern alternating current (AC) electricity supply system.**



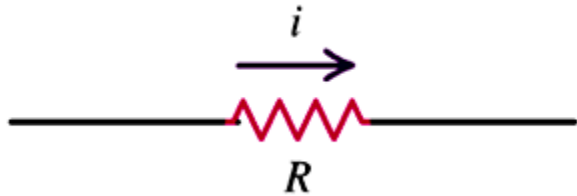
Lightning Master: Tesla in his lab

Wardenclyffe Tower facility and global wireless transmission

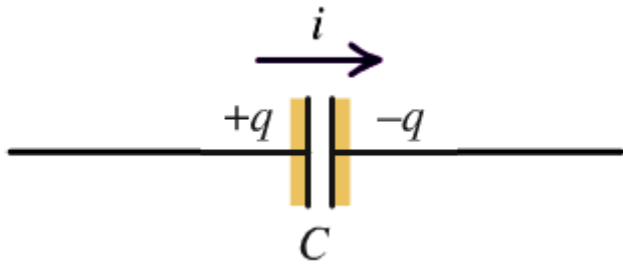


- ◆ LC Circuit
- ◆ LRC Circuit
- ◆ Power in AC Circuits
- ◆ Transformers

Ch31 EM oscillation & AC current

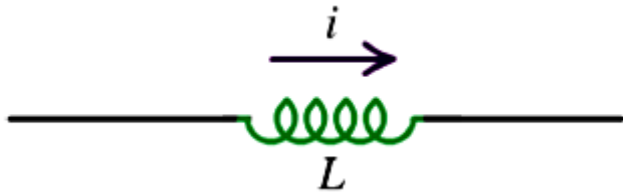


Energy is dissipated



Energy is stored in ***E***-field

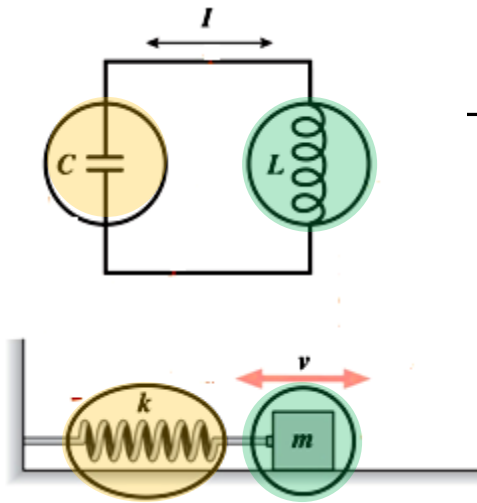
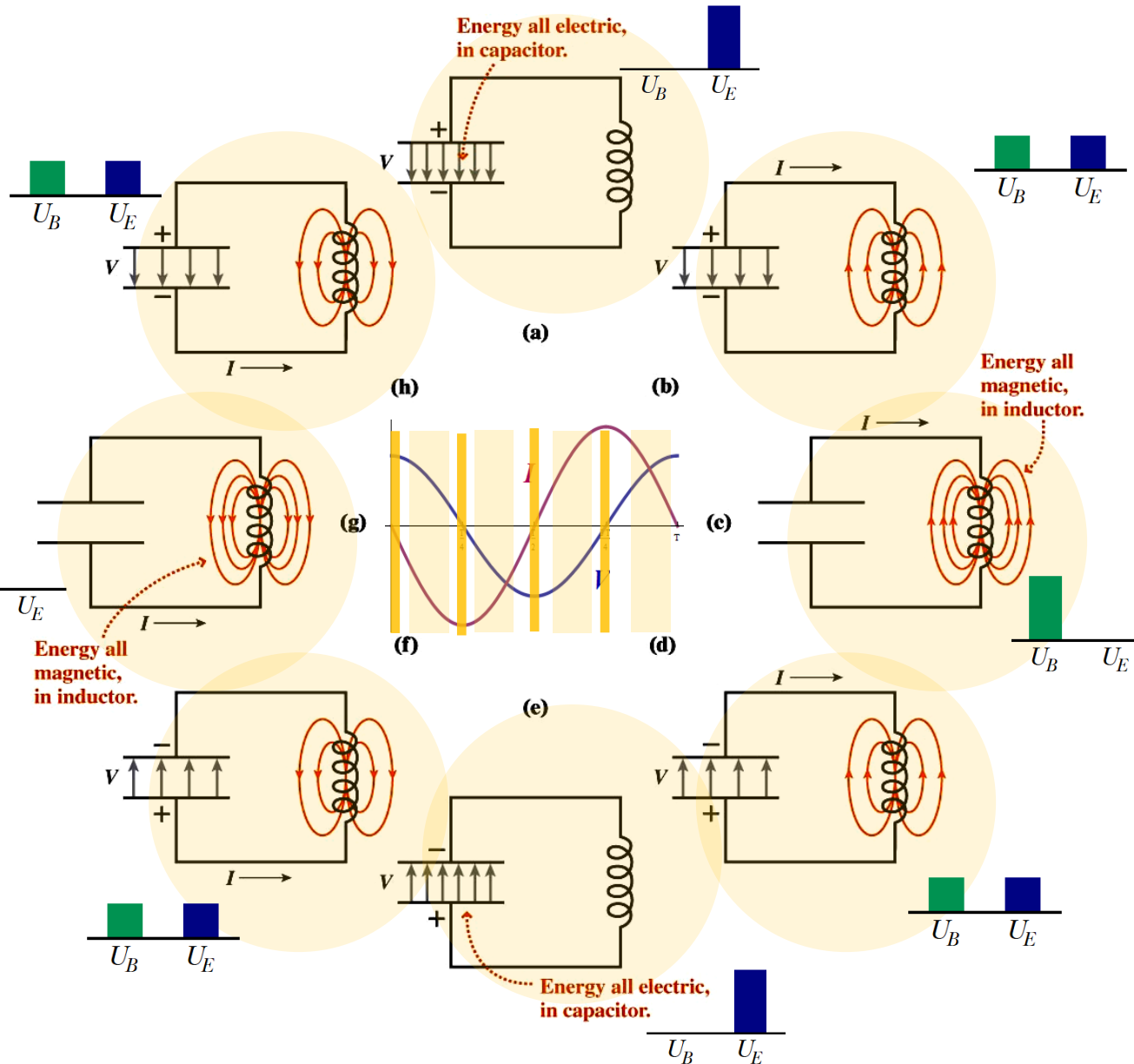
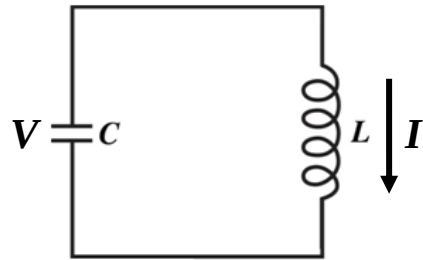
$$U_E = \frac{q^2}{2C}$$



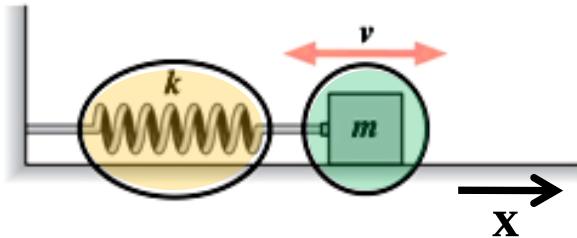
Energy is stored in ***B***-field

$$U_B = \frac{Li^2}{2}$$

LC Circuits



Mass-spring system VS LC oscillator



Mass-spring system

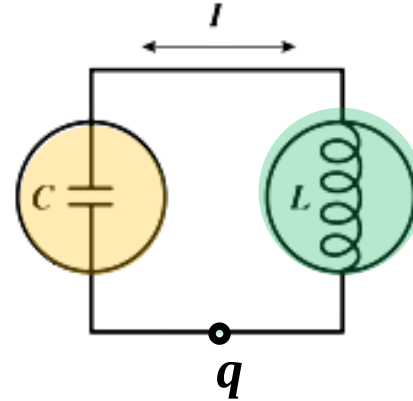
Spring : potential energy $\frac{1}{2}kx^2$

Mass : kinetic energy $\frac{1}{2}mv^2$

$$v = dx/dt$$

Angular frequency

$$\omega = \sqrt{\frac{k}{m}}$$



LC oscillator

Capacitor : electrical energy $\frac{1}{2}(1/C)q^2$

Inductor : magnetic energy $\frac{1}{2}Li^2$

$$i = dq/dt$$

$$\omega = \frac{1}{\sqrt{LC}}$$

LC oscillator

$$U = U_B + U_E = \text{constant}$$

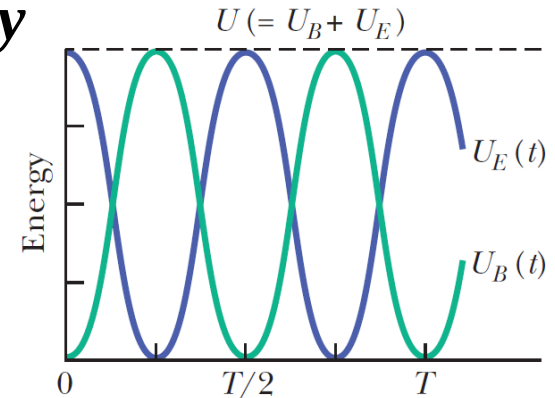
$$= \frac{Li^2}{2} + \frac{q^2}{2C}$$

$$\rightarrow \frac{dU}{dt} = 0$$

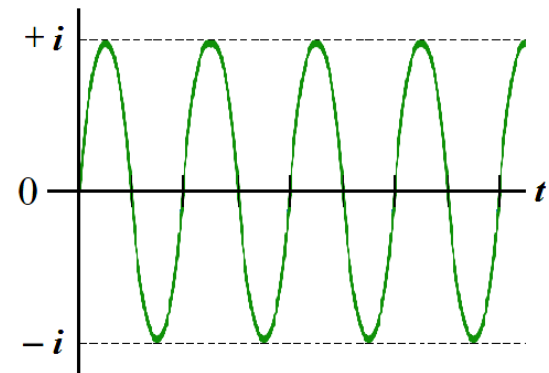
$$\cancel{L} i \frac{di}{dt} + \frac{q}{C} \frac{dq}{dt} = 0$$

$$L \frac{d^2q}{dt^2} + \frac{1}{C} q = 0$$

Energy

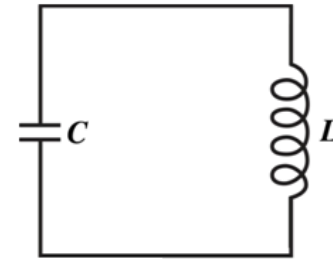


current



LC oscillator

$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = 0$$



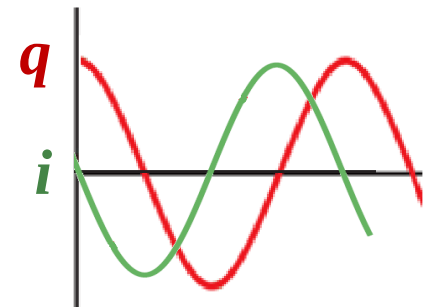
Solution for this 2nd order differential equation :

$$q = Q \cos(\omega t + \phi) \qquad \omega = \frac{1}{\sqrt{LC}}$$

q : instantaneous value

Q : amplitude

Check 1) plug this into the equation
2) the current and energy
3) how about the charge and current ?

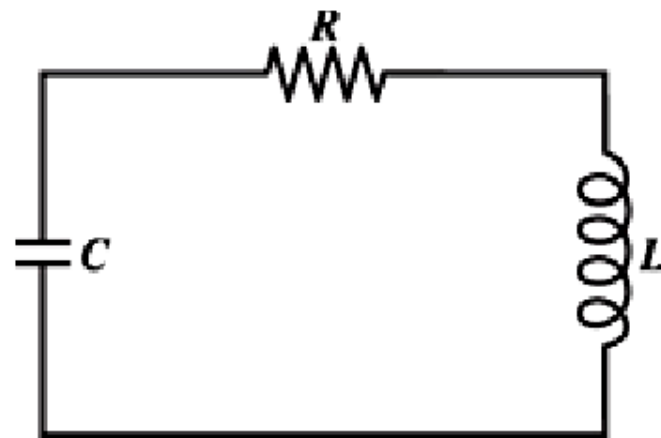


Damped oscillation in LRC circuit

$$U = \frac{Li^2}{2} + \frac{q^2}{2C} \quad \frac{dU}{dt} = -i^2 R$$

$$L \cancel{i} \frac{di}{dt} + \frac{\cancel{q}}{C} \frac{dq}{dt} = -\cancel{i}^2 R$$

$$L \frac{di}{dt} + i R + \frac{q}{C} = 0$$



$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$$

Damped oscillation in LRC circuit

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$$

Solution for this 2nd order differential equation :

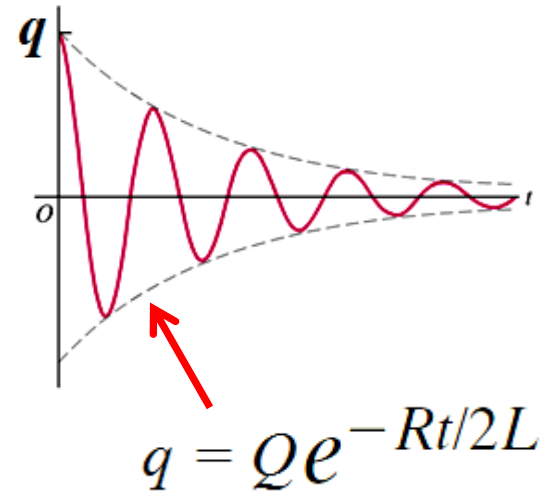
$$q = Q e^{-Rt/2L} \cos(\omega' t + \phi)$$

decaying **oscillating**

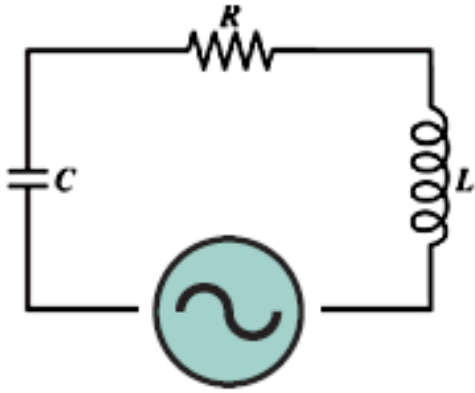
where

$$\omega' = \sqrt{\omega^2 - (R/2L)^2} \qquad \omega = \frac{1}{\sqrt{LC}}$$

$$\text{If } R \rightarrow 0, \omega' \rightarrow \omega$$



Forced oscillation in LRC circuit



If an external alternating current (AC) is connected, we call it **driven oscillation / forced oscillation**

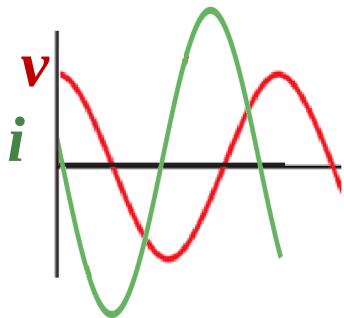
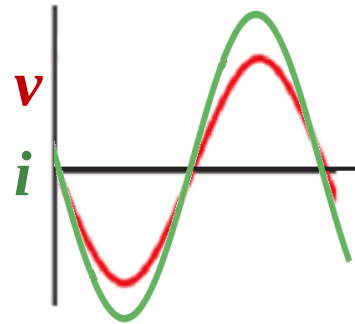
$$v = V_{max} \cos \omega_d t$$

$$2\pi f_d = \omega_d$$

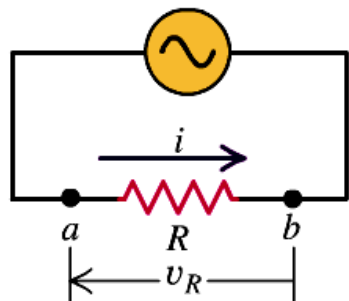
$$i = I_{max} \cos (\omega_d t - \phi)$$



**ϕ is the phase difference
 $\therefore v$ may not in phase
with i**

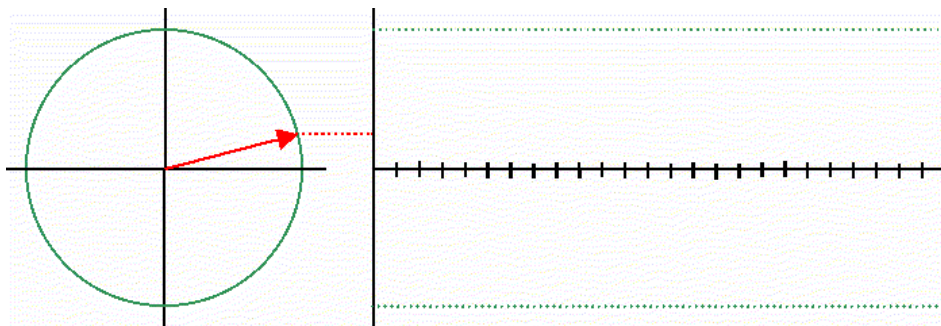


Alternating emf and a resistor

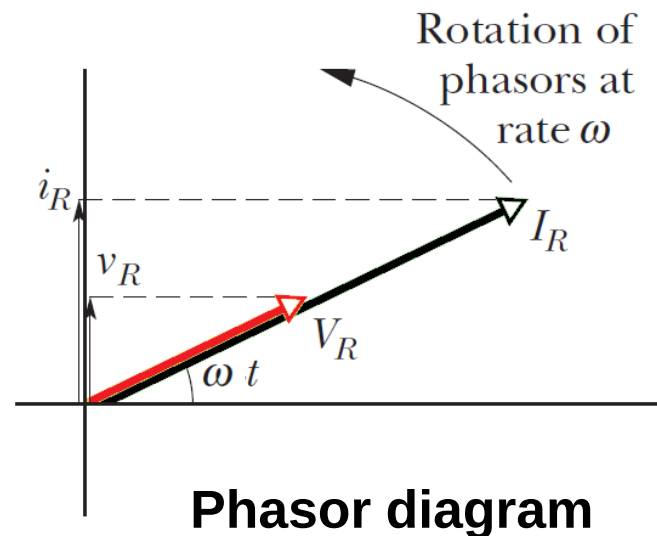
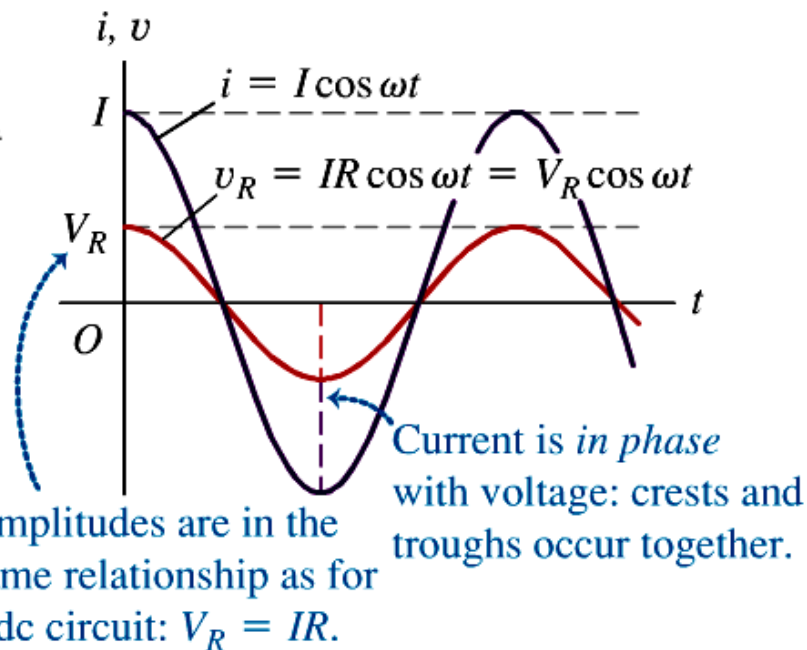
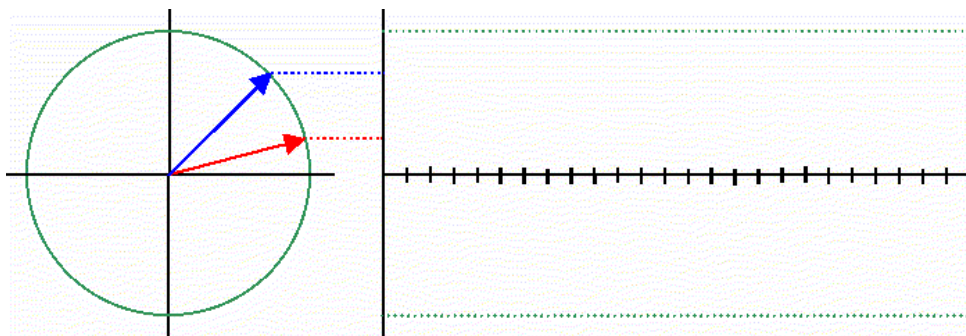


$$i = I \cos \omega t$$

$$v_R = iR = (IR) \cos \omega t$$

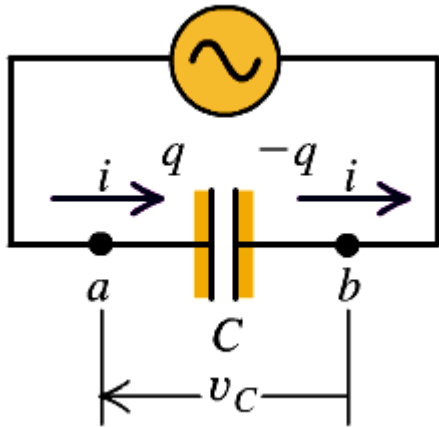


Phasor diagram



Phasor diagram

Alternating emf and a capacitor



$$i = \frac{dq}{dt} = I \cos \omega t$$

$$q = \frac{I}{\omega} \sin \omega t$$

$$\therefore q = C v_C$$

$$\therefore v_C = \frac{I}{\omega C} \sin \omega t$$

$$v_C = \frac{I}{\omega C} \cos(\omega t - 90^\circ)$$

voltage **lags** (落后) current by 90° or,
current **leads** (领先) voltage by 90°

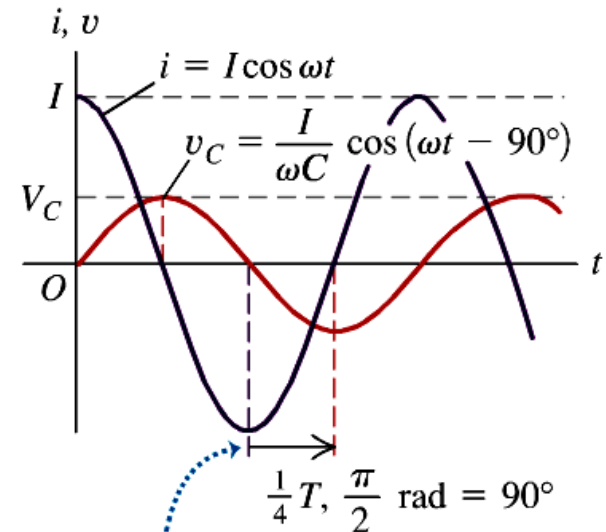
$$\text{maximum voltage } V_C = \frac{I}{\omega C}$$

(voltage amplitude)

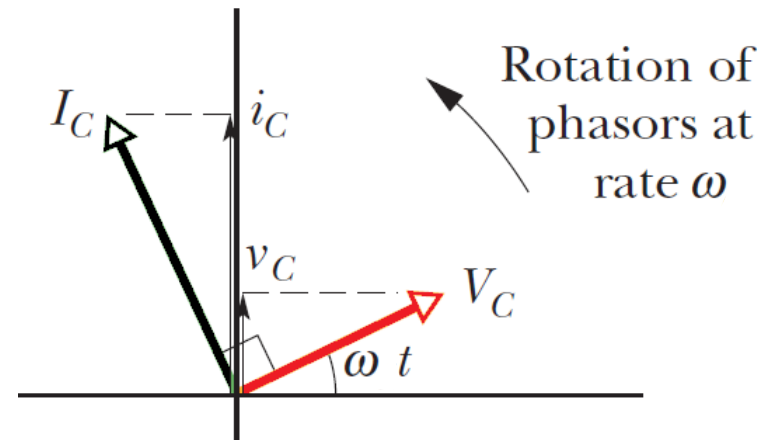
Similar to $V = IR$, we define :

$$X_C = \frac{1}{\omega C} \quad (\text{capacitive reactance})$$

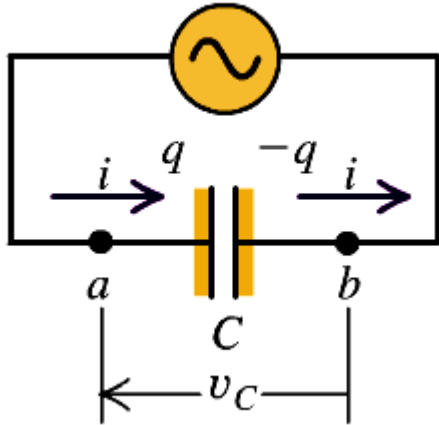
SI unit of X_C is Ω



Voltage curve *lags* current curve by a quarter-cycle (corresponding to $\phi = -\pi/2 \text{ rad} = -90^\circ$).



Alternating emf and a capacitor



$$i = \frac{dq}{dt} = I \cos \omega t \quad q = \frac{I}{\omega} \sin \omega t$$

$$\therefore q = Cv_C \quad \therefore v_C = \frac{I}{\omega C} \sin \omega t$$

$$v_C = \frac{I}{\omega C} \cos(\omega t - 90^\circ)$$

voltage **lags** (落后) current by 90°
current **leads** (领先) voltage by 90°

$$\text{maximum voltage } V_C = \frac{I}{\omega C} \quad (\text{voltage amplitude})$$

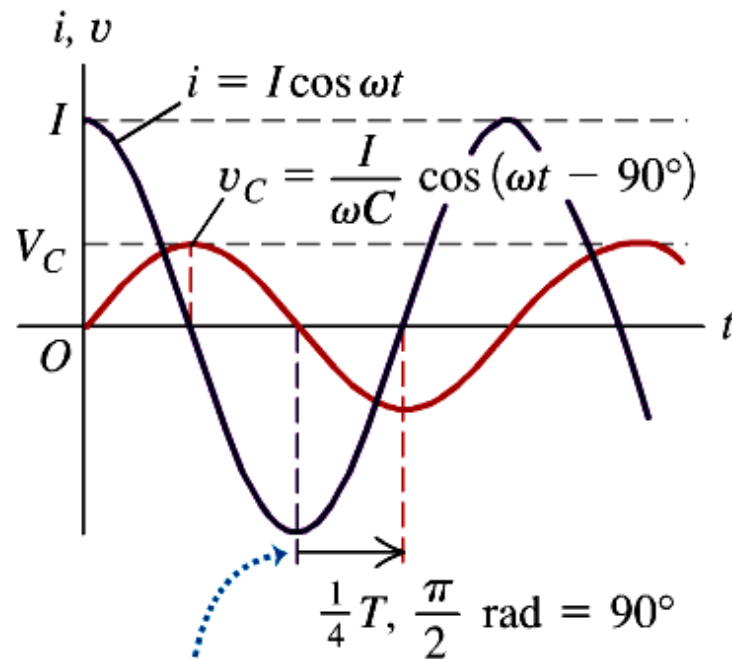
Similar to $V = IR$, we define (容抗):

$$X_C = \frac{1}{\omega C} \quad (\text{capacitive reactance})$$

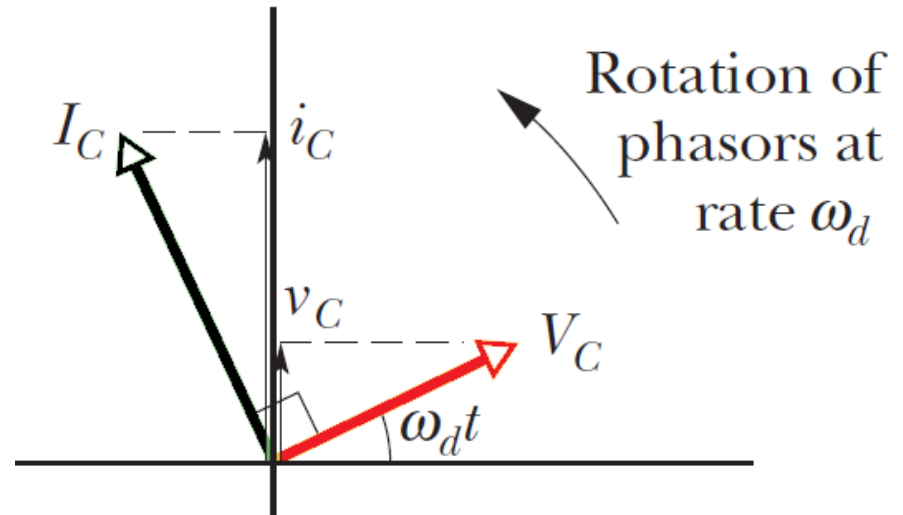
SI unit of X_C is Ω

Alternating emf and a capacitor

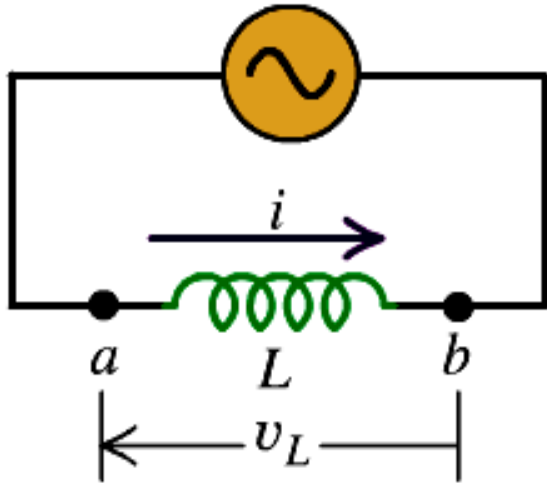
$$v_C = \frac{I}{\omega C} \cos(\omega t - 90^\circ)$$



Voltage curve *lags* current curve by a quarter-cycle (corresponding to $\phi = -\pi/2$ rad = -90°).



Alternating emf and an inductor



$$v_L = L \frac{di}{dt} \qquad i = I \cos \omega t$$
$$= L \frac{d}{dt} (I \cos \omega t) = -I\omega L \sin \omega t$$

$$v_L = I\omega L \cos(\omega t + 90^\circ)$$

voltage **leads** (领先) current by 90°

current **lags** (落后) voltage by 90°

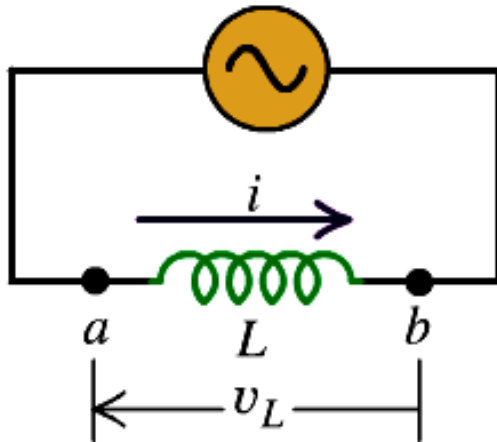
maximum voltage $V_L = I\omega L$ (voltage amplitude)

Similar to $V = IR$, we define (感抗):

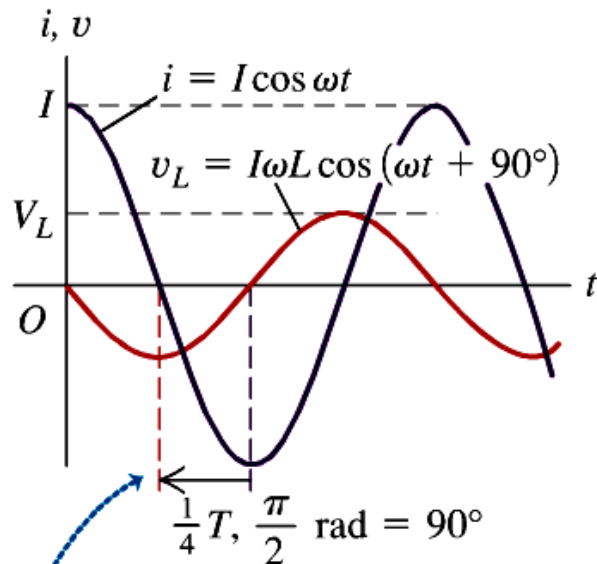
$$X_L = \omega L \text{ (inductive reactance)}$$

SI unit of X_L is Ω

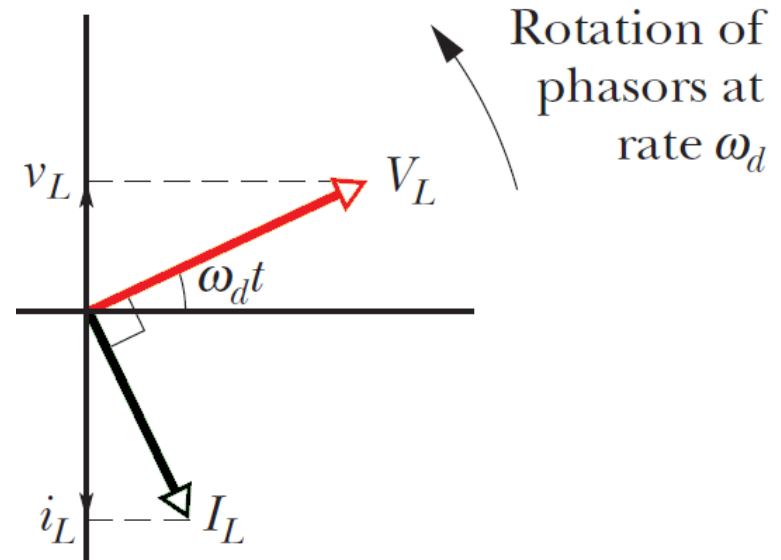
Alternating emf and an inductor



$$v_L = I\omega L \cos(\omega t + 90^\circ)$$



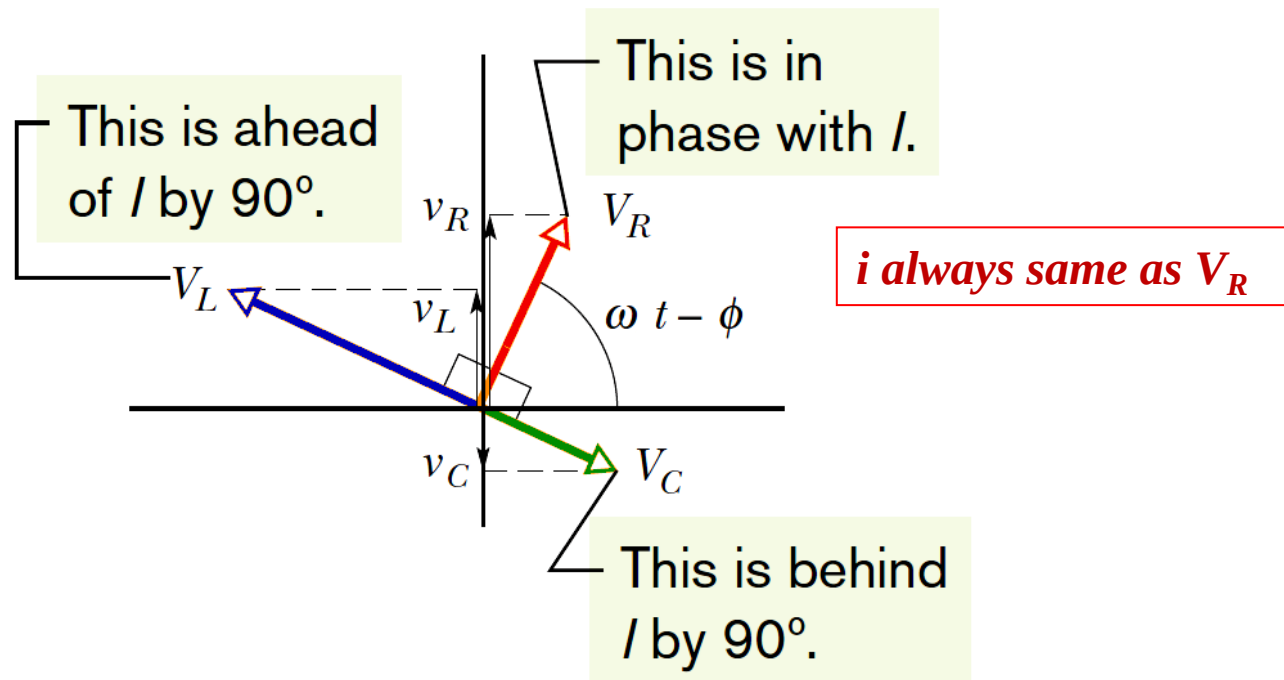
Voltage curve *leads* current curve by a quarter-cycle (corresponding to $\phi = \pi/2$ rad = 90°).



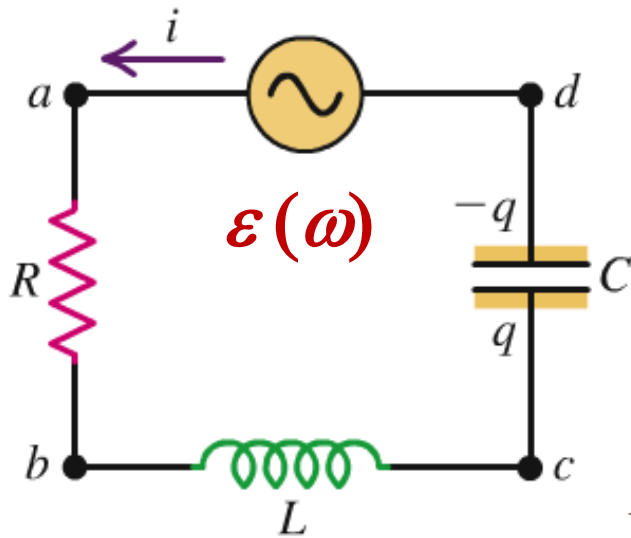
Resistor, Inductor, and Capacitor

Circuit Elements with Alternating Current

Circuit Element	Amplitude Relationship	Circuit Quantity	Phase of v
Resistor	$V_R = IR$	R	In phase with i
Inductor	$V_L = IX_L$	$X_L = \omega L$	Leads i by 90°
Capacitor	$V_C = IX_C$	$X_C = 1/\omega C$	Lags i by 90°



Series RLC circuit



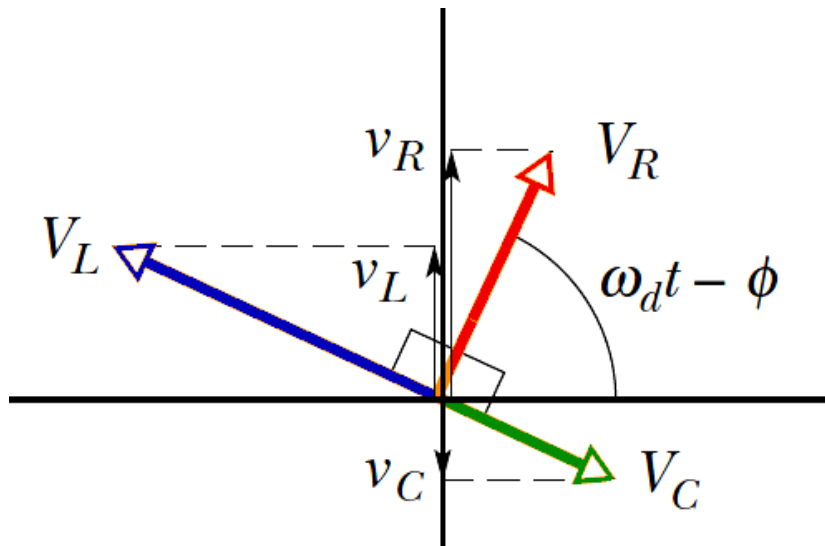
$$V_R^2 + (V_L - V_C)^2 = \epsilon_m^2$$

$$(IR)^2 + (IX_L - IX_C)^2 = \epsilon_m^2$$

$$I = \frac{\epsilon_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$X_L = \omega L$$

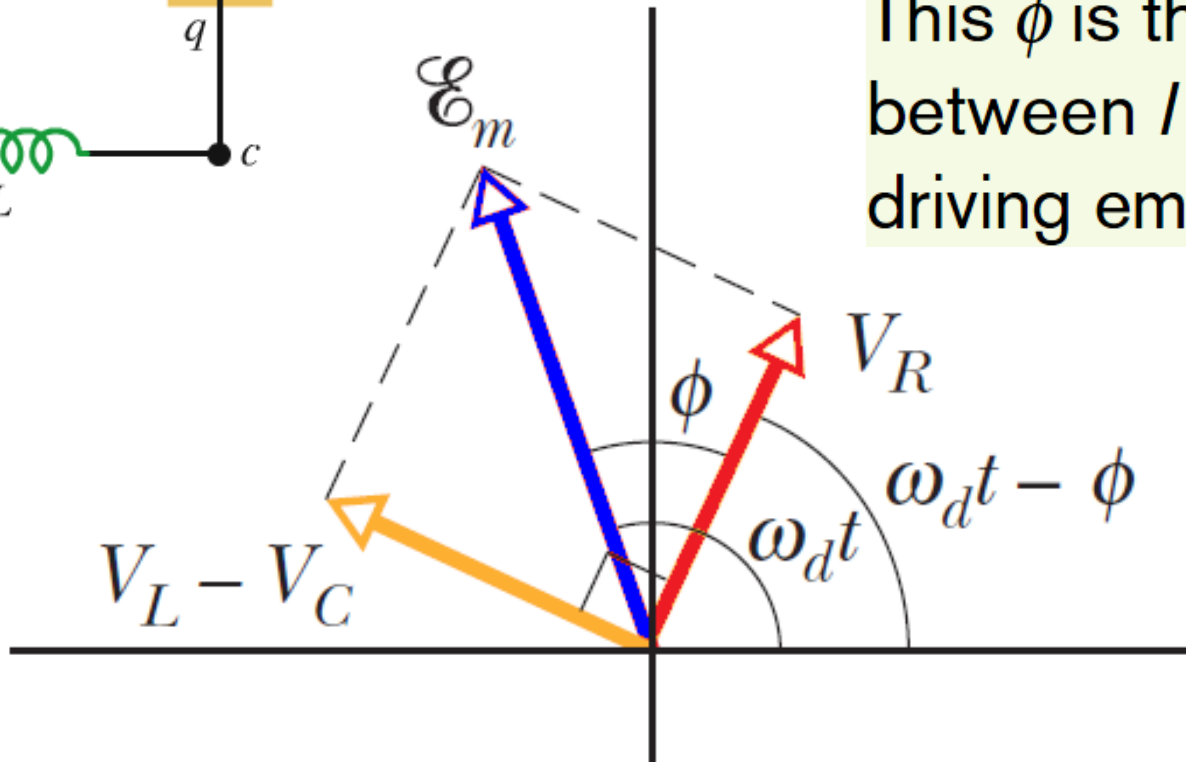
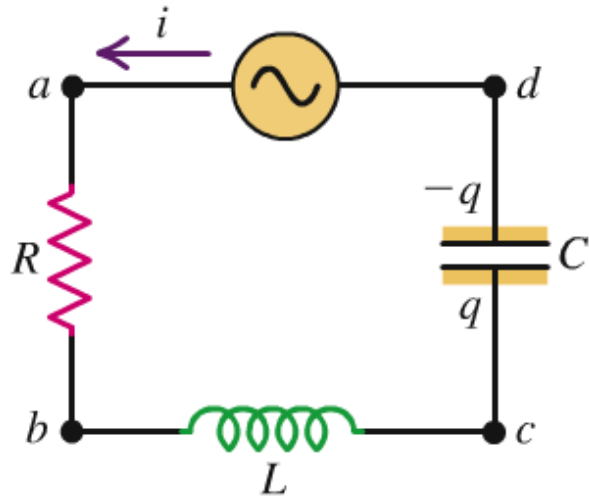
$$X_C = \frac{1}{\omega C}$$



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Z : Impedance (阻抗)

Series RLC circuit



This ϕ is the angle between i and the driving emf.

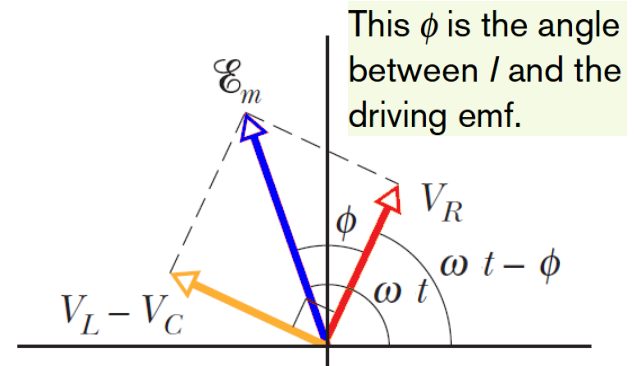
i always same as V_R

The phase constant

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{IX_L - IX_C}{IR} = \frac{X_L - X_C}{R}$$

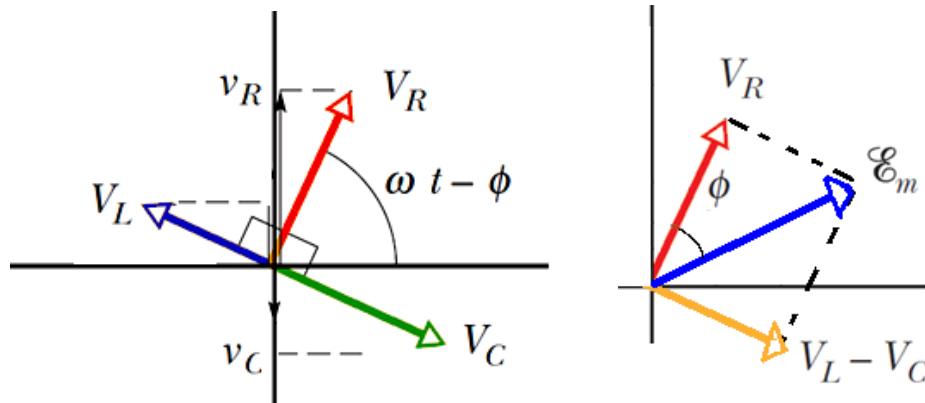
$$X_L = \omega L$$

$$X_C = \frac{1}{\omega C}$$

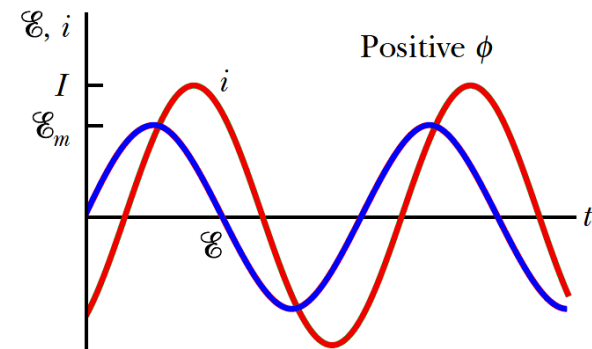
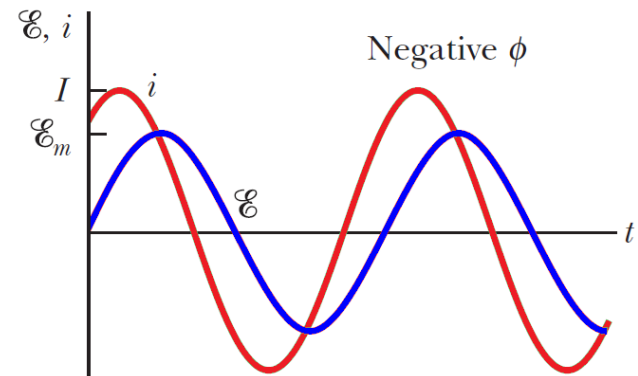


i always same as V_R

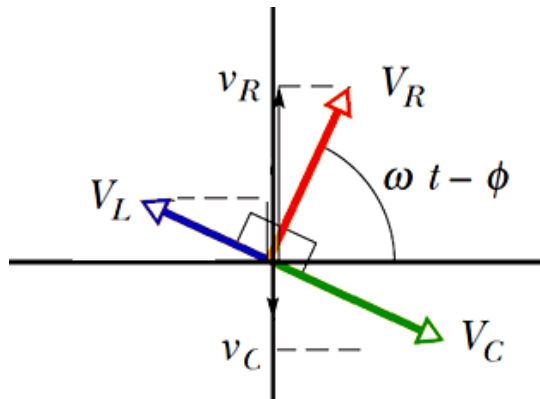
$X_C > X_L$ The circuit is more capacitive



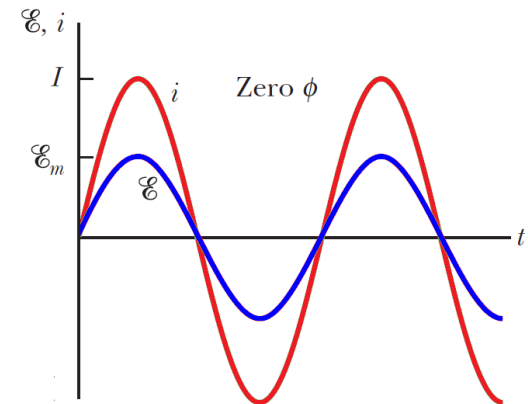
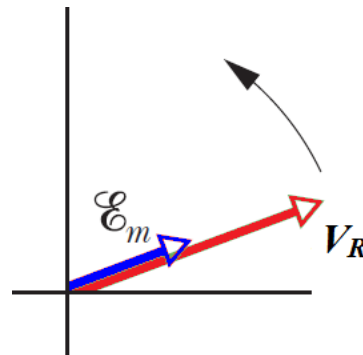
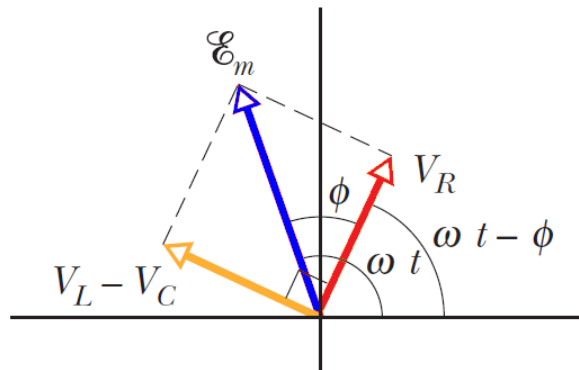
$X_L > X_C$ The circuit is more inductive



Resonance



What if $X_C = X_L$?



$$I = \frac{\mathcal{E}_m}{\sqrt{R^2 + \cancel{(X_L - X_C)^2}}}$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

The circuit is in resonance!

-
- LC Circuit
 - LRC Circuit
 - Power in AC Circuits
 - Transformers

Power in AC circuits

In DC circuit : $\text{Power} = i^2 R$

In AC circuit : $i = I \sin(\omega t - \phi)$

$$\begin{aligned}\text{Power} = i^2 R &= [I \sin(\omega t - \phi)]^2 R \\ &= I^2 R \sin^2(\omega t - \phi)\end{aligned}$$

$$\begin{aligned}\text{Average power} = P_{\text{avg}} &= \frac{I^2 R}{2} \\ &= \left(\frac{I}{\sqrt{2}} \right)^2 R\end{aligned}$$

 **Root-mean-square (rms) value of i**

$$I_{\text{rms}} = \frac{I}{\sqrt{2}} \text{ rms current}$$

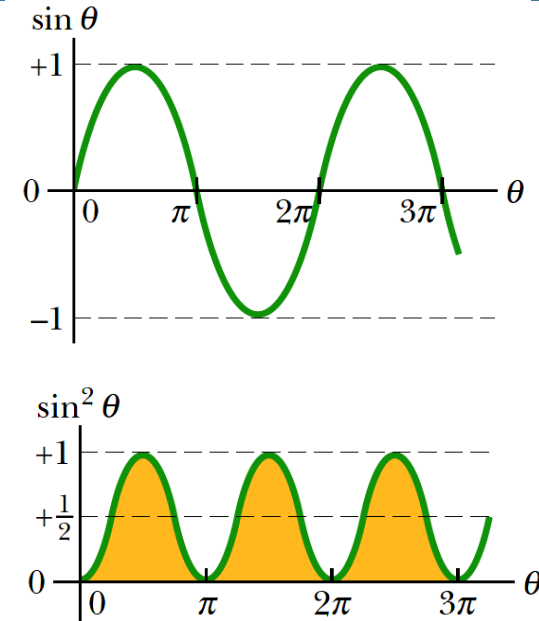
$$\therefore P_{\text{avg}} = I_{\text{rms}}^2 R \text{ (average power)}$$

rms voltage

$$V_{\text{rms}} = \frac{V}{\sqrt{2}}$$

rms emf

$$\mathcal{E}_{\text{rms}} = \frac{\mathcal{E}_m}{\sqrt{2}}$$

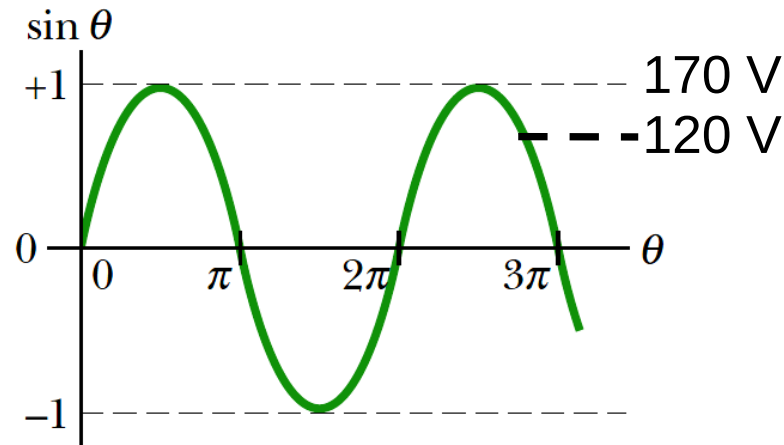


Measuring AC voltage/current by multimeter

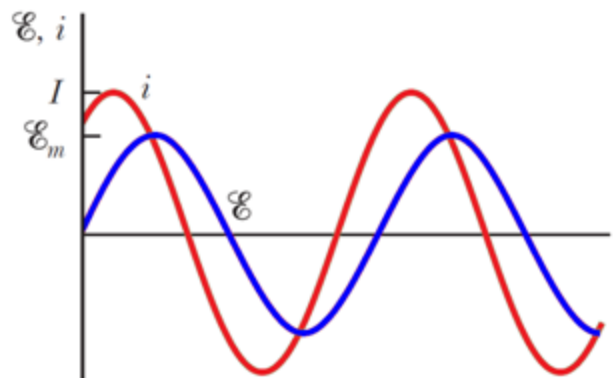
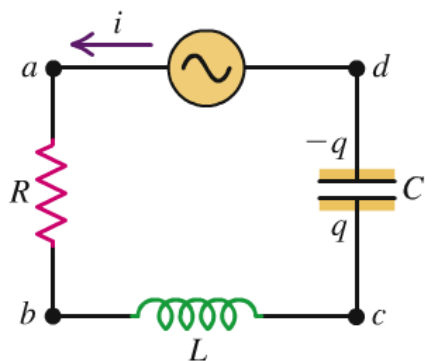


The value shown in multimeter is rms value

If it shows a reading of 120 V for an AC voltage signal, the peak value is $(120 \div \sqrt{2}) = 170 \text{ V}$



Power in AC circuits with resistor, inductor, capacitor



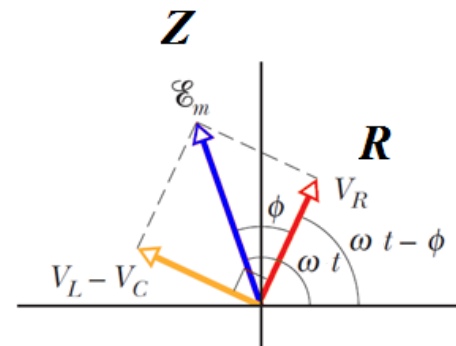
Energy is stored in the E-field in the capacitor, B-field in the inductor, and dissipated in the resistor

$$I_{\text{rms}} = \frac{\mathcal{E}_{\text{rms}}}{Z} = \frac{\mathcal{E}_{\text{rms}}}{\sqrt{R^2 + (X_L - X_C)^2}} \quad ?$$

$$P_{\text{avg}} = I_{\text{rms}}^2 R = \frac{\mathcal{E}_{\text{rms}}}{Z} I_{\text{rms}} R = \mathcal{E}_{\text{rms}} I_{\text{rms}} \frac{R}{Z}$$

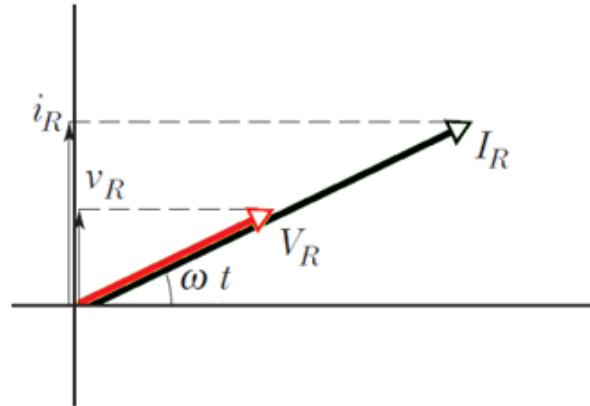
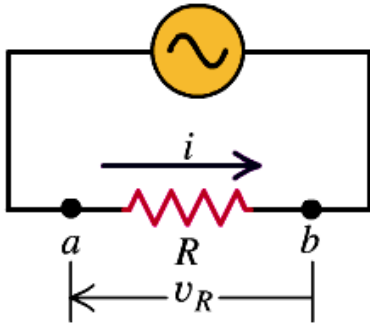
$$P_{\text{avg}} = \mathcal{E}_{\text{rms}} I_{\text{rms}} \cos \phi$$

Power factor

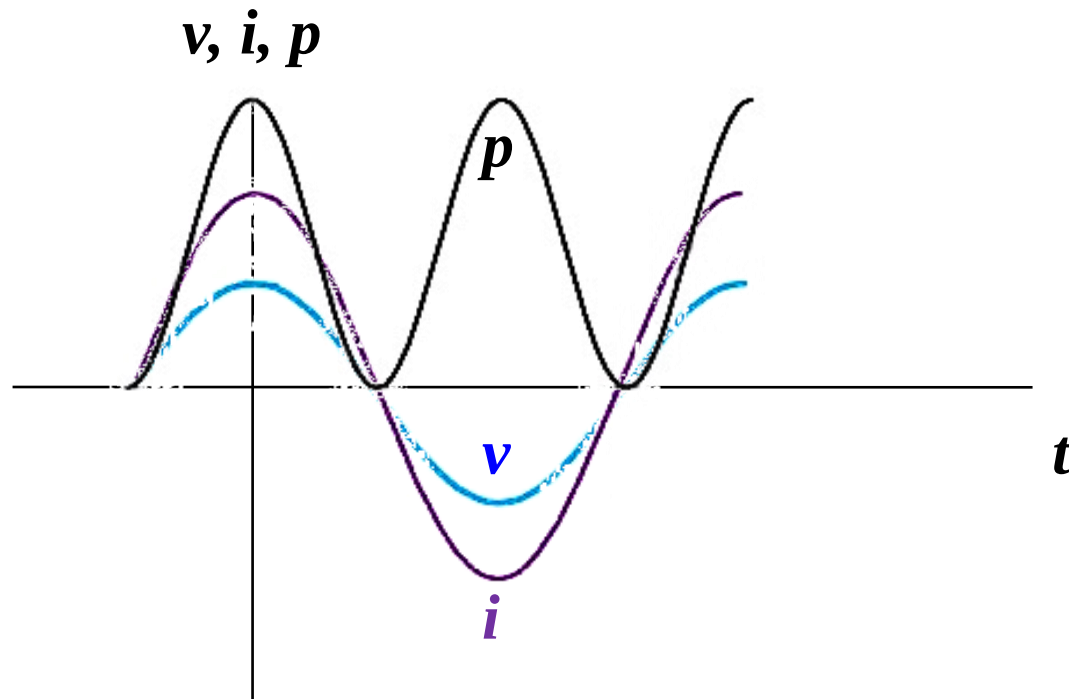


$$\cos \phi = \frac{V_R}{\mathcal{E}_m} = \frac{IR}{IZ} = \frac{R}{Z}$$

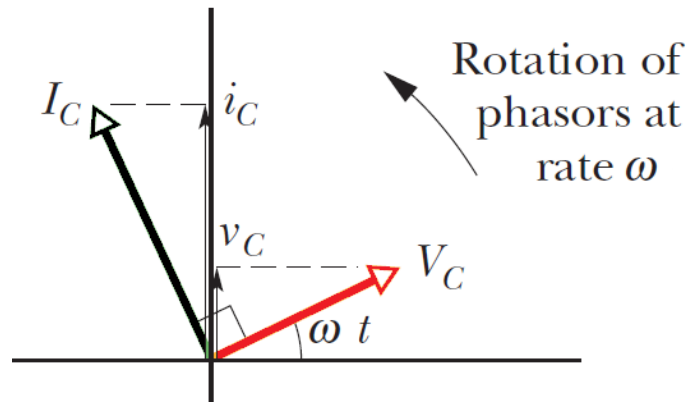
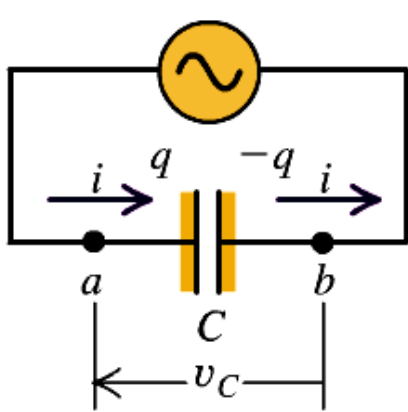
More on power in AC - resistor



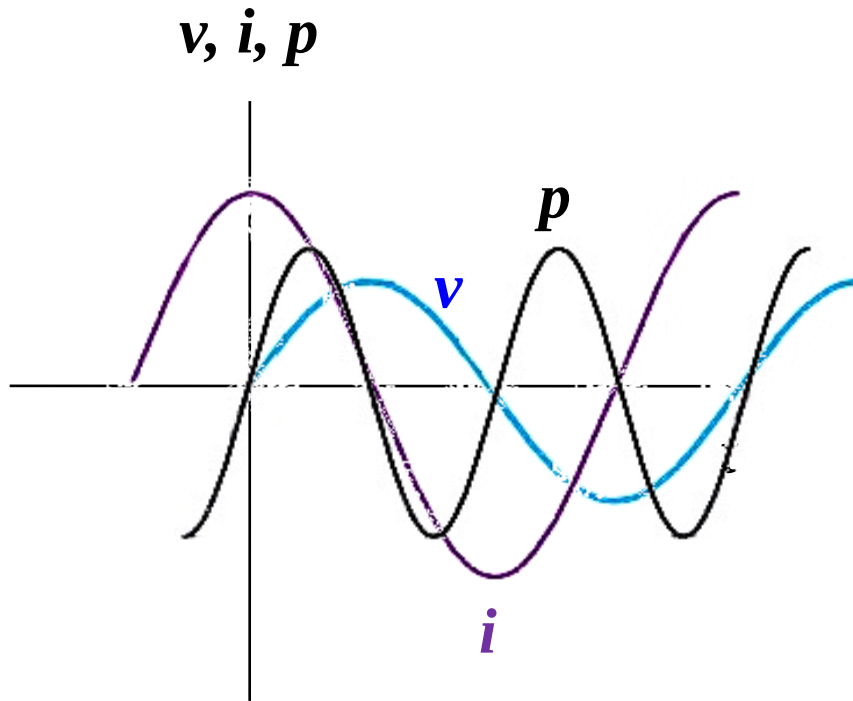
$$P_{\text{avg}} = \mathcal{E}_{\text{rms}} I_{\text{rms}} \cos \phi$$



More on power in AC - capacitor

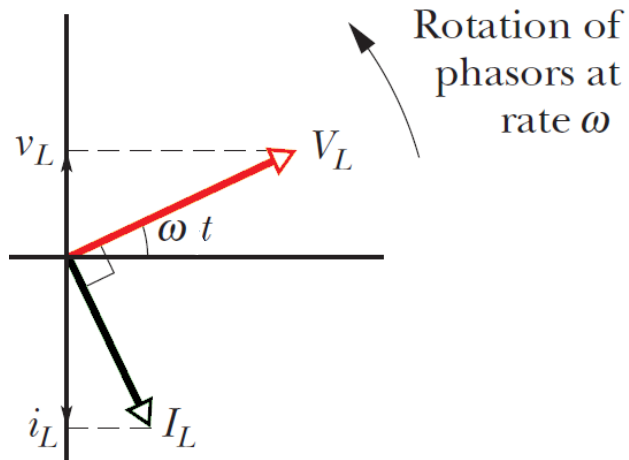
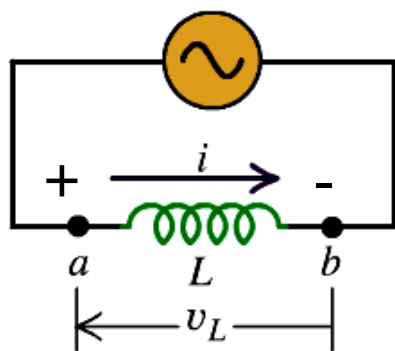


$$P_{\text{avg}} = \mathcal{E}_{\text{rms}} I_{\text{rms}} \cos \phi$$



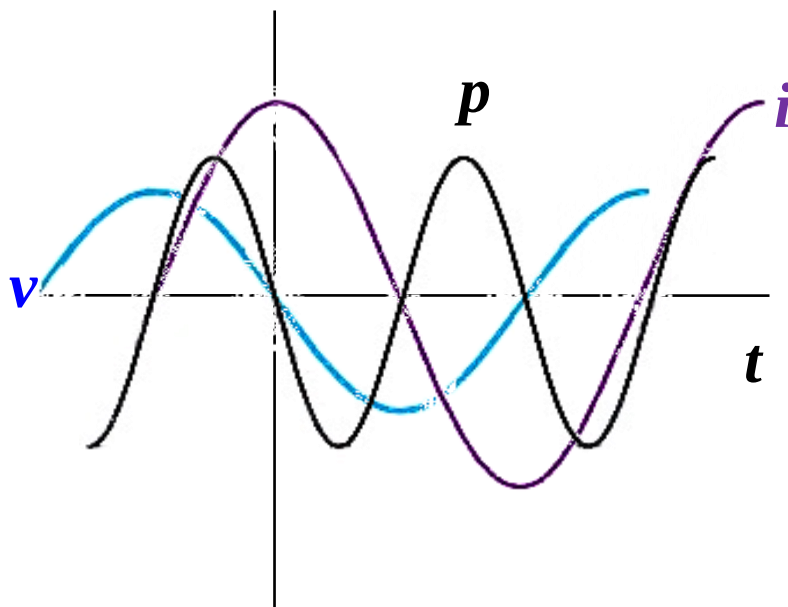
Average power = 0

More on power in AC - inductor



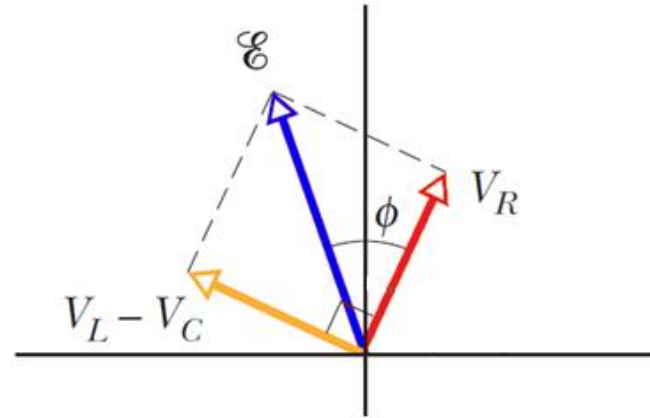
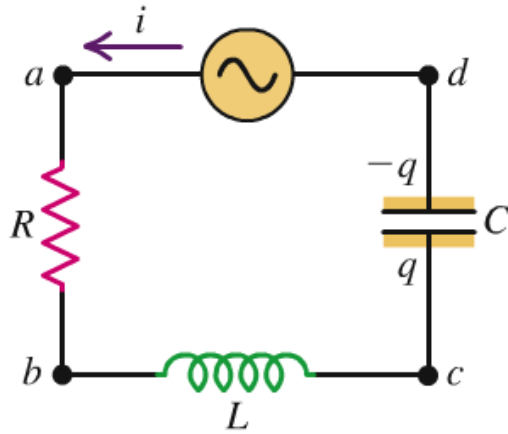
$$P_{\text{avg}} = \mathcal{E}_{\text{rms}} I_{\text{rms}} \cos \phi$$

v, i, p

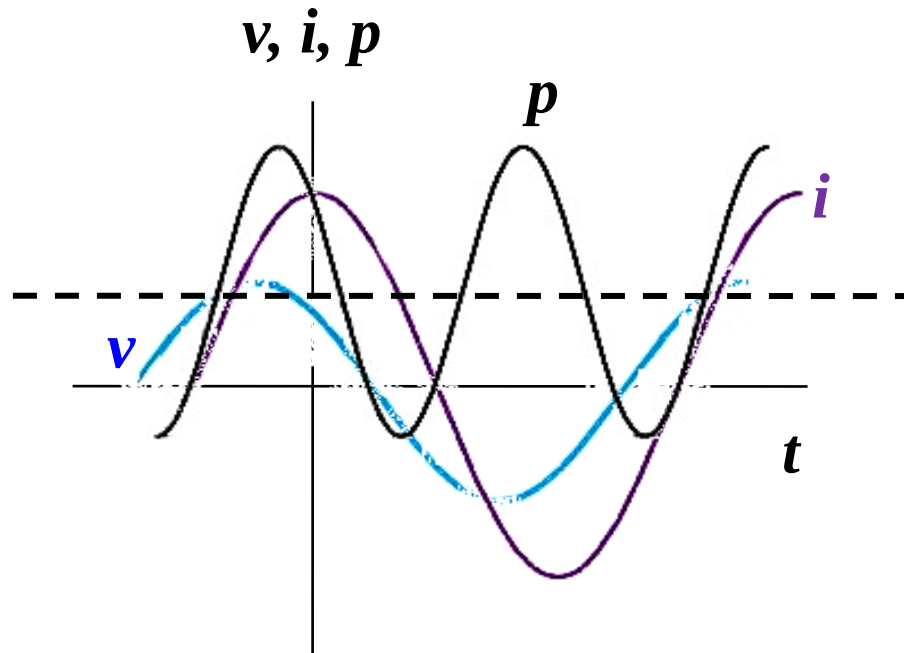


Average power = 0

More on power in AC – arbitrary ac circuit

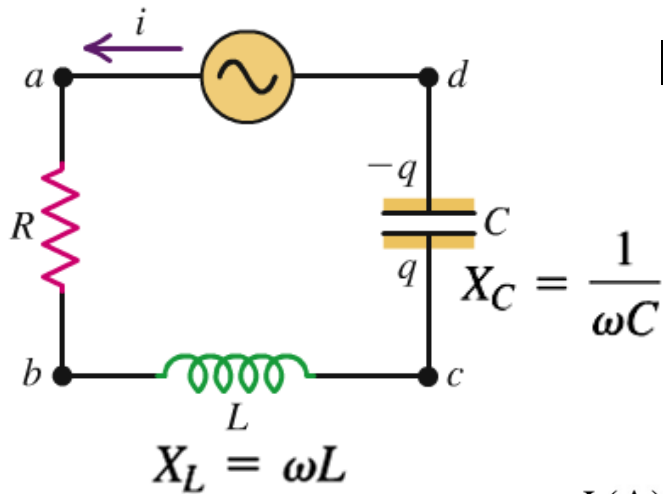


$$P_{\text{avg}} = \mathcal{E}_{\text{rms}} I_{\text{rms}} \cos \phi$$



Average power $\neq 0$

Resonance curves



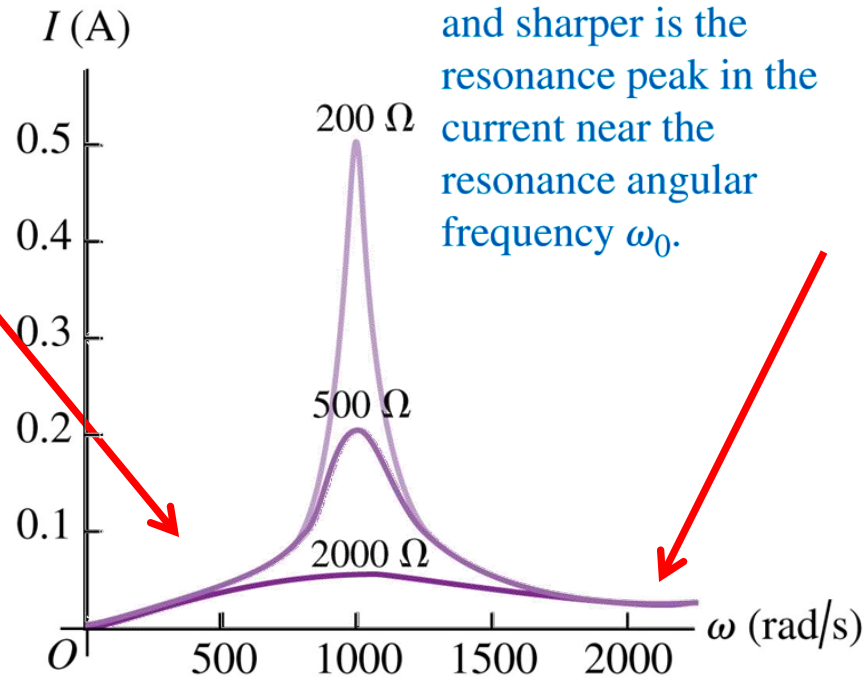
Impedance

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

resonance: $\omega = \frac{1}{\sqrt{LC}}$

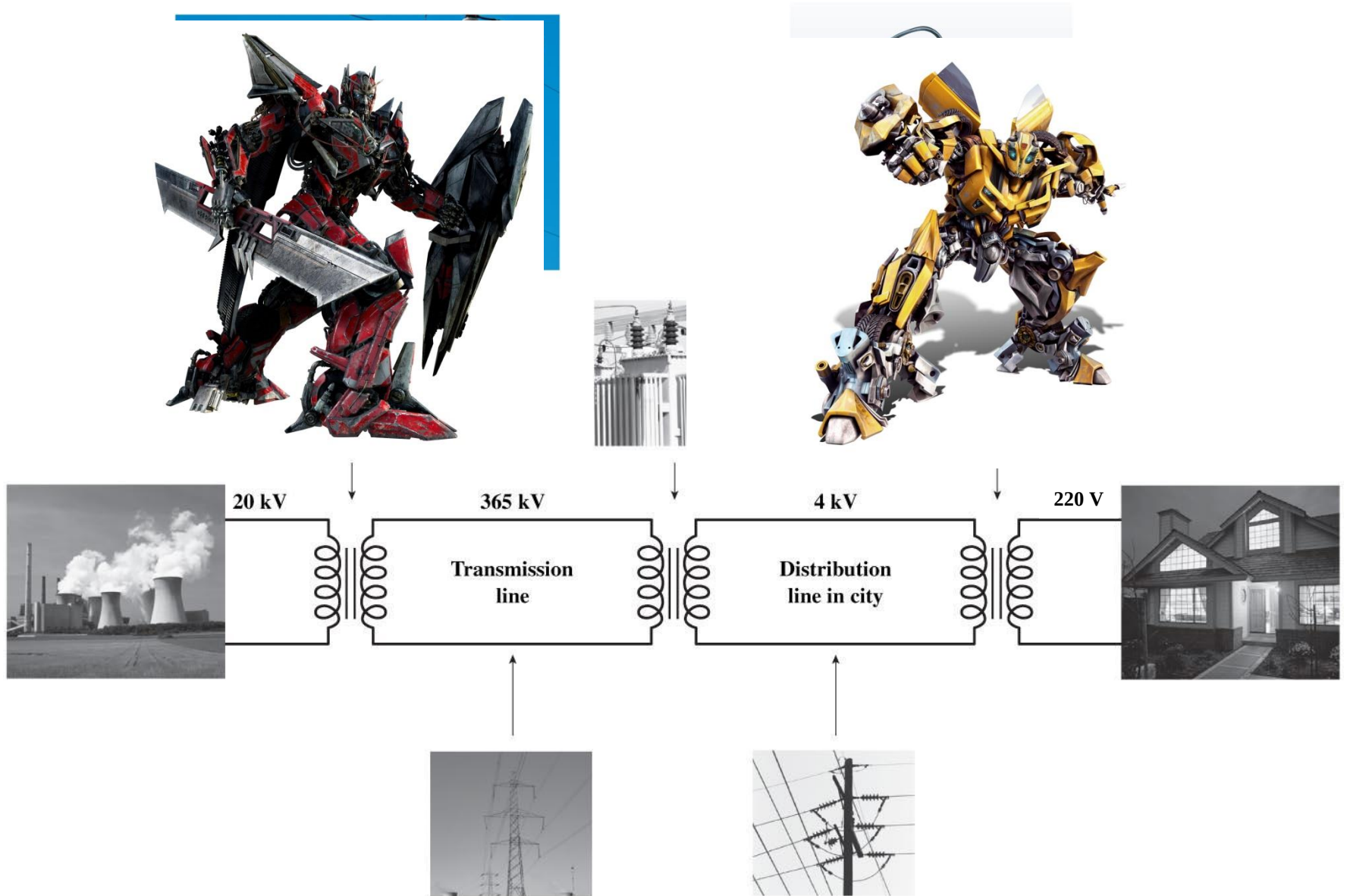
The lower a circuit's resistance, the higher and sharper is the resonance peak in the current near the resonance angular frequency ω_0 .

X_C is greater
more capacitive



X_L is greater
more inductive

Transformers



Transformers

$$\mathcal{E}_1 = -N_1 \frac{d\Phi_B}{dt}$$

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_B}{dt}$$

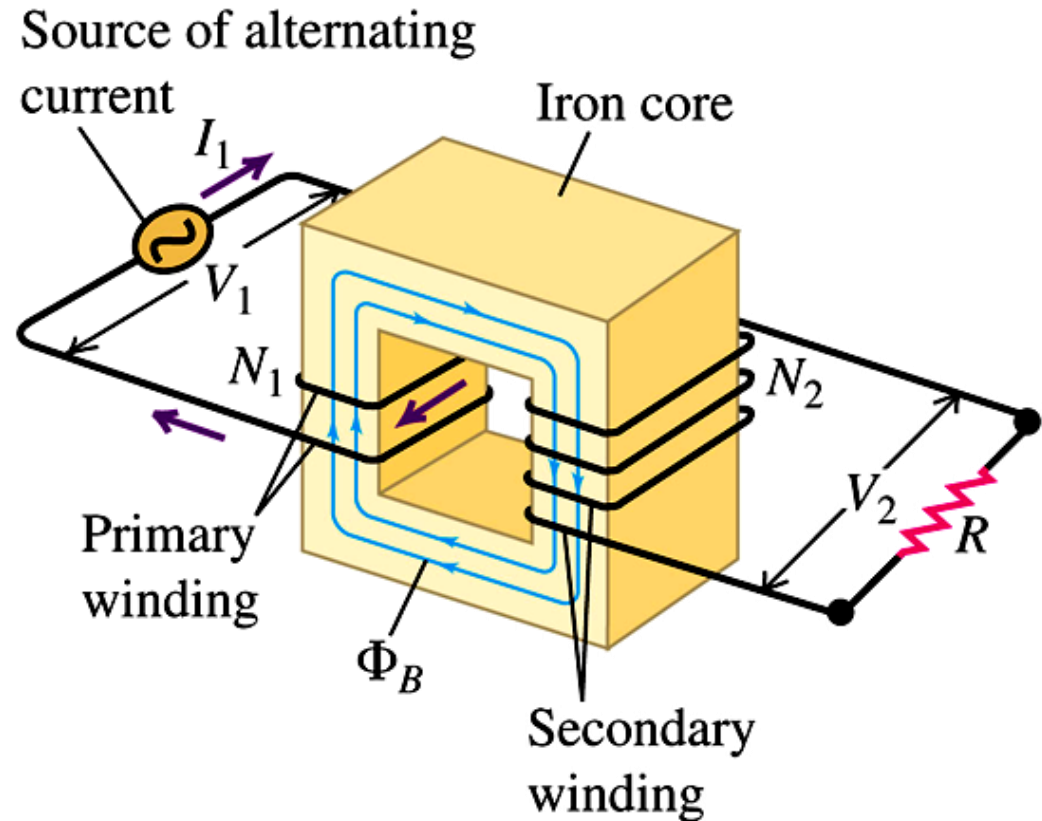
Φ_B is the same for each turn of the primary and secondary windings

$$\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

If $N_2 > N_1$, a step-up transformer

If $N_2 < N_1$, a step-down transformer



Transformers

By conservation of energy :

$$V_1 I_1 = V_2 I_2$$

put in:

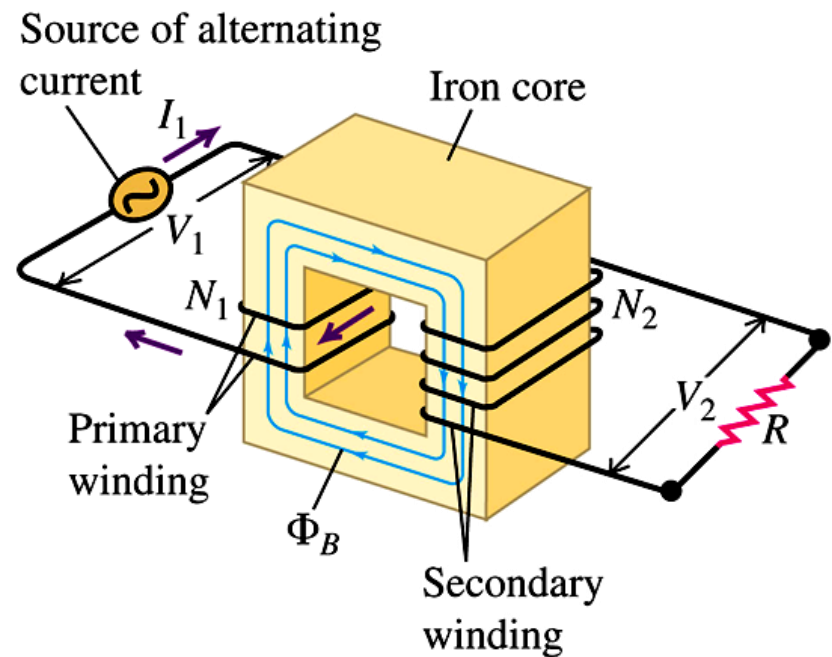
$$V_2 = V_1 \frac{N_2}{N_1}$$

and

$$I_2 = V_2 / R$$

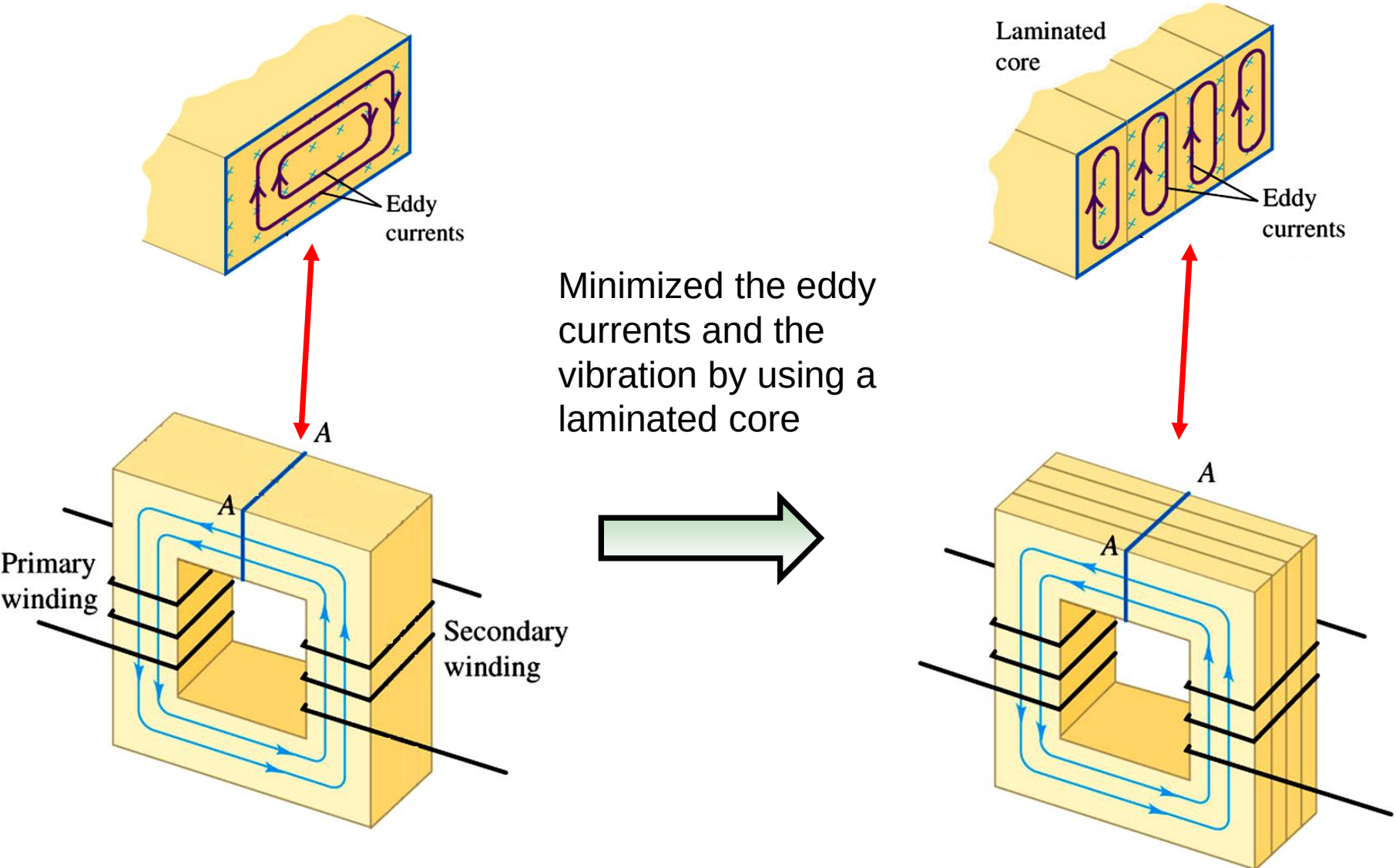
we have

$$\frac{V_1}{I_1} = \frac{R}{(N_2/N_1)^2} >$$



**The value of load resistance (负载电阻)
seen by the generator**

Energy loss by eddy currents



Why AC not DC?

